

PRAKTIKUM SISTEM OPERASI

MODUL IV

(MEMBACA SOURCE CODE XINU)



Oleh:

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(2311104065)

PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

UNIVERSITAS TELKOM

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JURNAL

1. Apa nama image yang dihasilkan setelah melakukan kompilasi pada Xinu? Berapa ukuran file tersebut? Ada pada folder apa file image tersebut?

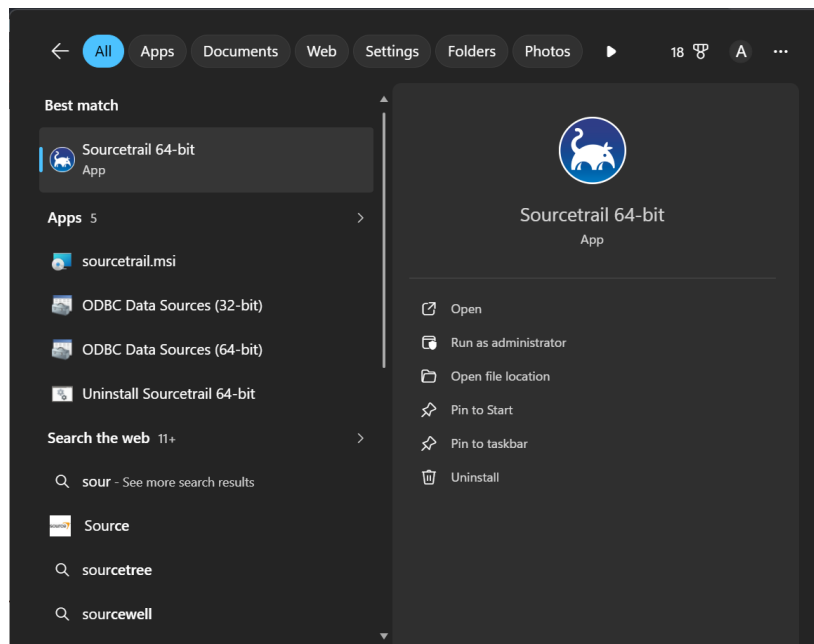
Jawab:

file image hasil kompilasi Xinu bernama xinu.elf, Ukuran file biasanya berkisar antara 500 KB – 2 MB tergantung konfigurasi dan fitur yang dikompilasi, berada di folder ./compile (direktori xinu/compile/)

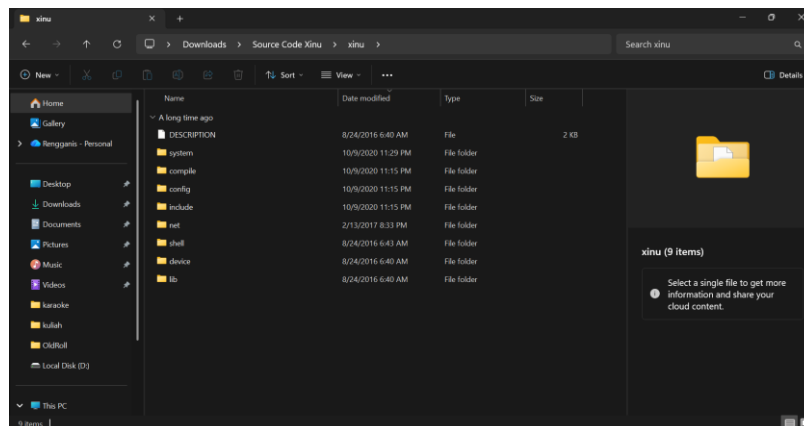
2. Membaca source code Xinu

- a. Download SourceTrail

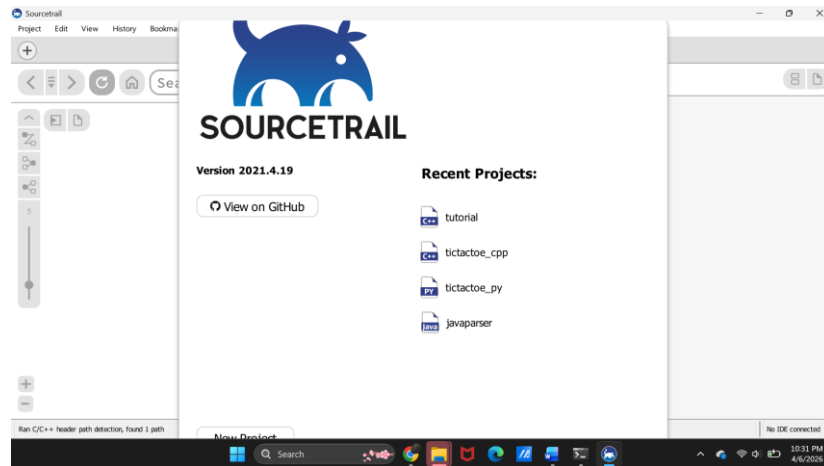
Jawab:



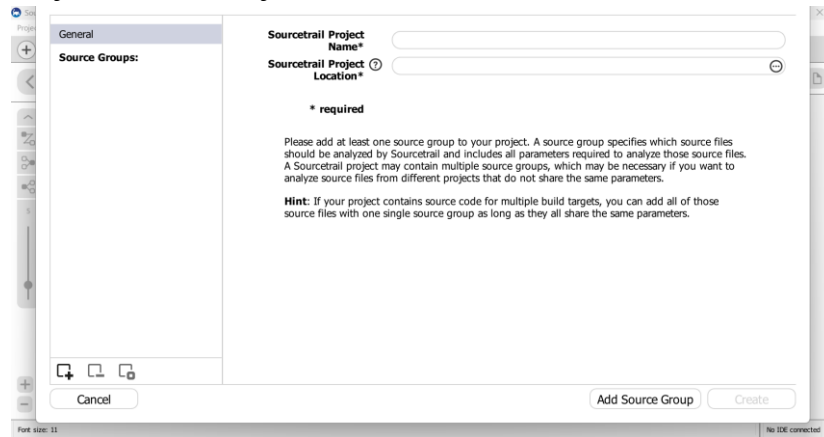
- b. Download file source code xinu



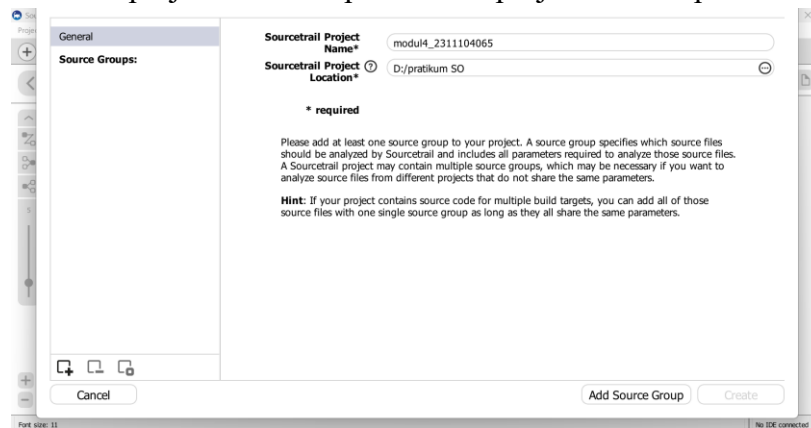
c. Jalankan SourceTrail



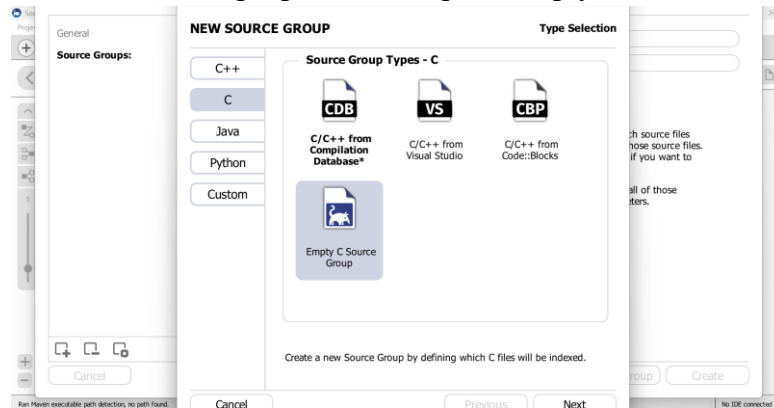
d. Project → New Project



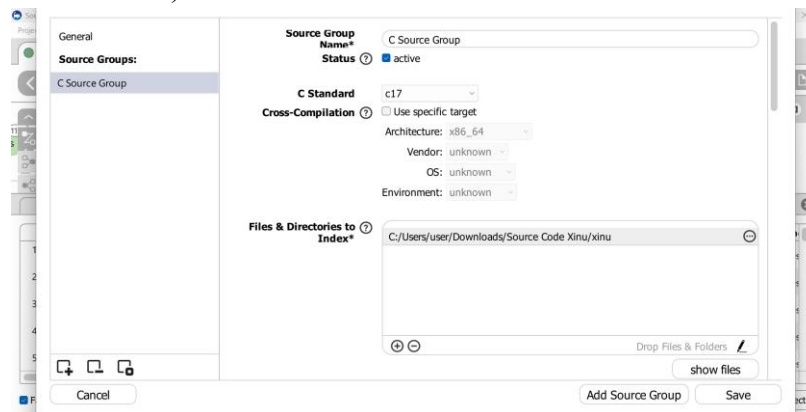
e. Isi nama project xinu dan pilih lokasi project di manapun



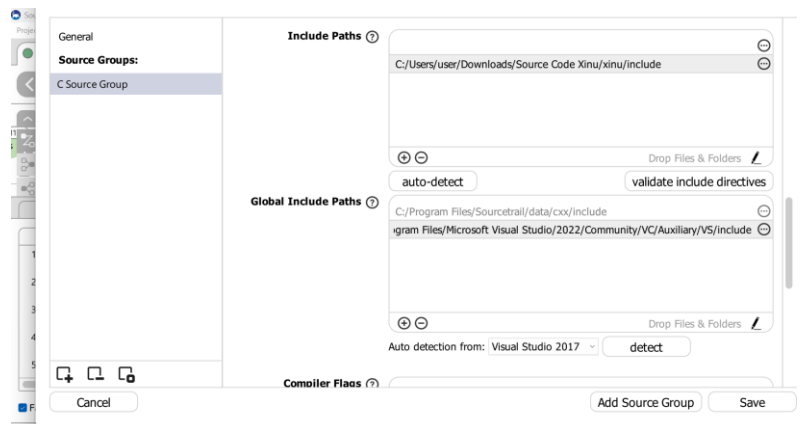
- f. Add Source Groups, pilih C, lalu pilih Empty C Source Group



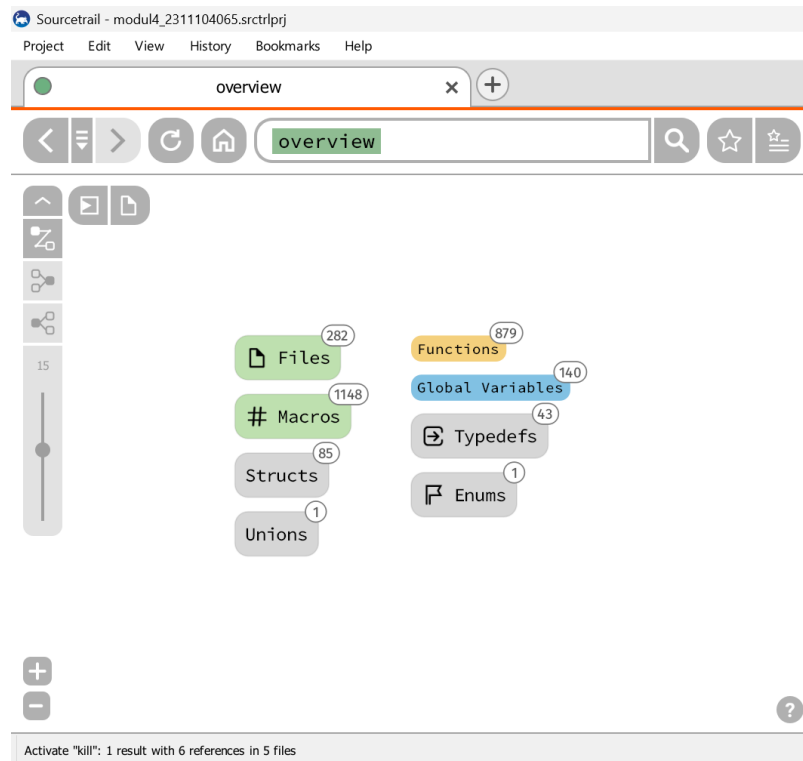
- g. File & Directories to Index: masukkan semua folder Xinu (yang sebelumnya telah di download)



- h. Include Paths: .../xinu/include

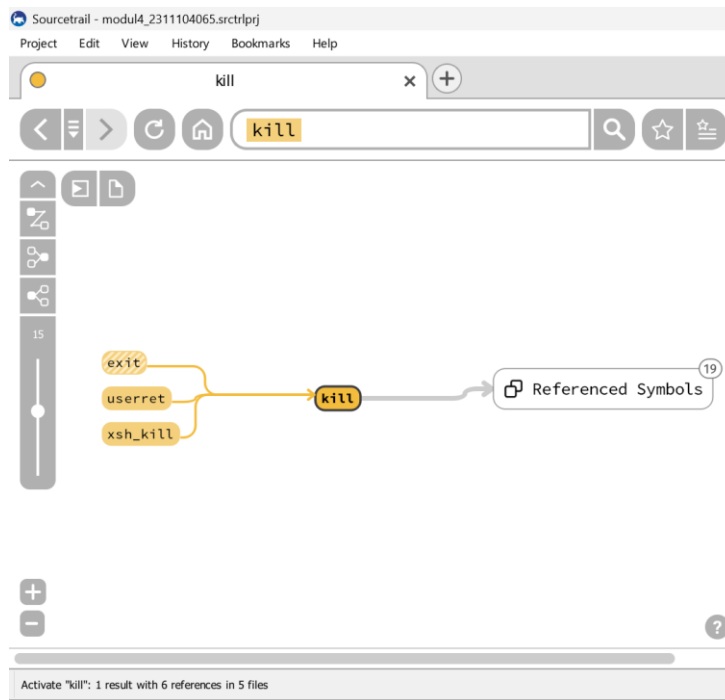


i. Create



j. Eksplorasi source code Xinu

- Kill



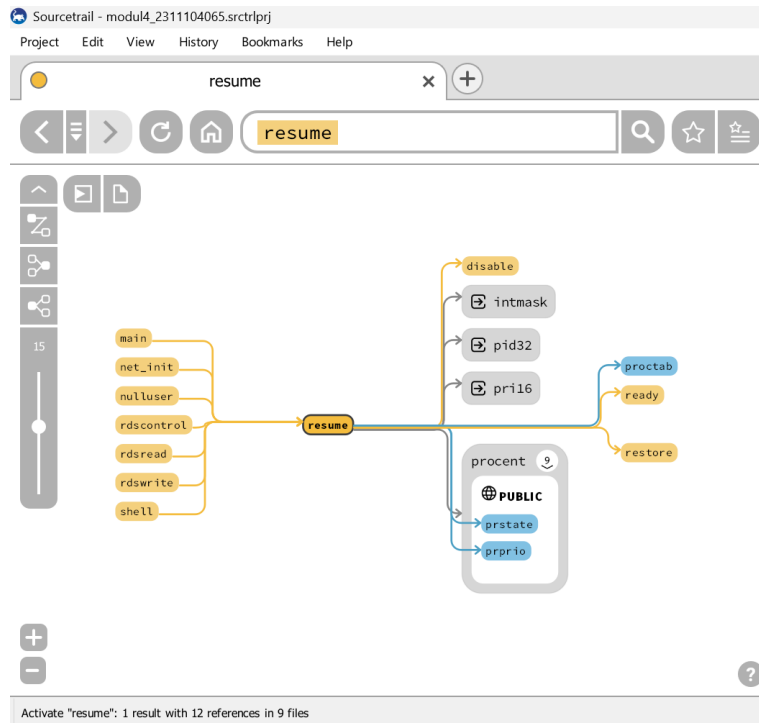
- Main

The screenshot shows the Sourcetrail application window titled "Sourcetrail - modul4_2311104065.srctrlprj". The menu bar includes "Project", "Edit", "View", "History", "Bookmarks", and "Help". The top toolbar shows navigation icons and a search bar containing "main". Below the toolbar, a vertical sidebar on the left contains icons for navigation and a scroll bar. The main workspace displays three yellow nodes: "main()", "main()", and "main(int, char * *)". At the bottom, a status bar indicates "Activate 'main': 3 results with 5 references in 3 files".

- Create

The screenshot shows the Sourcetrail application window titled "Sourcetrail - modul4_2311104065.srctrlprj". The menu bar includes "Project", "Edit", "View", "History", "Bookmarks", and "Help". The top toolbar shows navigation icons and a search bar containing "create". Below the toolbar, a vertical sidebar on the left contains icons for navigation and a scroll bar. The main workspace displays a diagram where five yellow nodes ("main", "net_init", "nulluser", "rdsinit", and "shell") are connected by lines to a central yellow node labeled "create". An arrow points from the "create" node to a box labeled "Referenced Symbols" with a circled number "14". At the bottom, a status bar indicates "Activate 'create': 1 result with 10 references in 7 files".

- Resume



3. Mencari struktur data dari proses pada Xinu OS. Struktur data proses ada pada file apa? Informasi apa saja yang disimpan dalam struktur data tersebut?

Jawab:

File process.h berlokasi di ~xinu/include/process.h

```
xinu@xinu-develop-end:~$ find / -name "process.h" 2>/dev/null
/home/xinu/prev-version/include/process.h
/home/xinu/xinu/include/process.h
/var/lib/dkms/virtualbox-guest/4.3.28/build/include/internal/process.h
/var/lib/dkms/virtualbox-guest/4.3.28/build/include/iprt/process.h
/usr/src/virtualbox-guest-4.3.28/include/internal/process.h
/usr/src/virtualbox-guest-4.3.28/include/iprt/process.h
xinu@xinu-develop-end:~$
```

Informasi yang disimpan meliputi

```
xinu@xinu-develop-end:~$ grep -r "procent\|prstate\|prprio\|prstk" /home/xinu/xinu/include/process.h
(proctab[(x)].prstate == PR_FREE))
#define NDESC 5 /* Must be odd to make procent 4N bytes */
struct procent {
    uint16 prstate; /* Process state: PR_CURR, etc. */
    pri16 prprio; /* Process priority */
    char *prstkptr; /* Saved stack pointer */
    char *prstkbase; /* Base of run time stack */
    uint32 prstklen; /* Stack length in bytes */
extern struct procent proctab[];
xinu@xinu-develop-end:~$ grep -r "struct procent" /home/xinu/xinu/include/process.h
struct procent { /* Entry in the process table */
xinu@xinu-develop-end:~$ cat /home/xinu/xinu/include/process.h
/* process.h - isbadpid */

/* Maximum number of processes in the system */

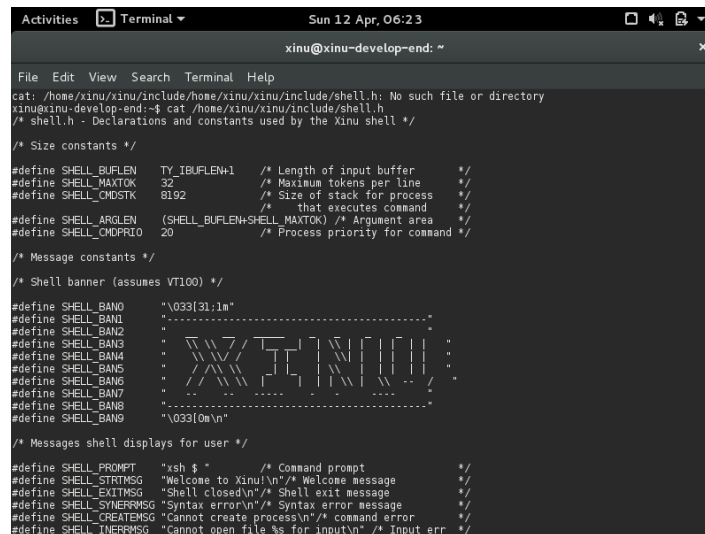
#ifndef NPROC
#define NPROC 8
#endif

/* Process state constants */

#define PR_FREE 0 /* Process table entry is unused */
```

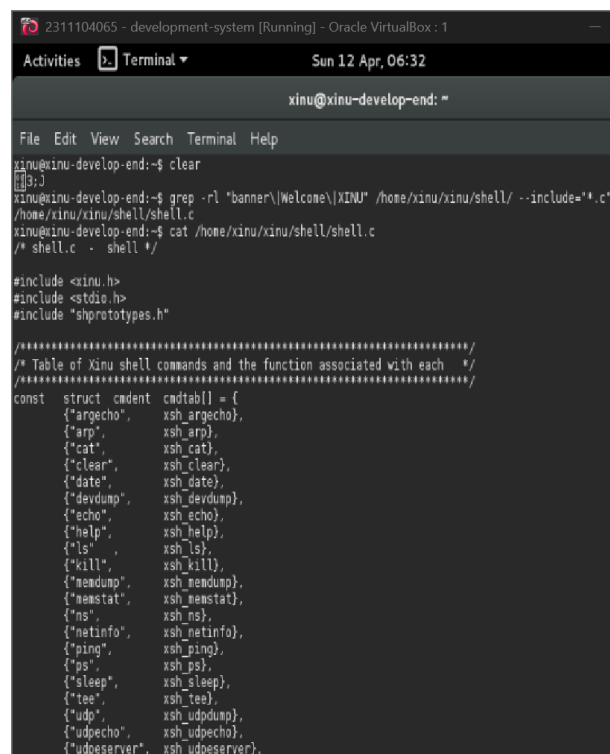
4. Mengubah welcome banner pada Xinu

- Carilah file yang menyimpan banner Xinu! Hint: file berekstensi .h pada direktori xinu/include



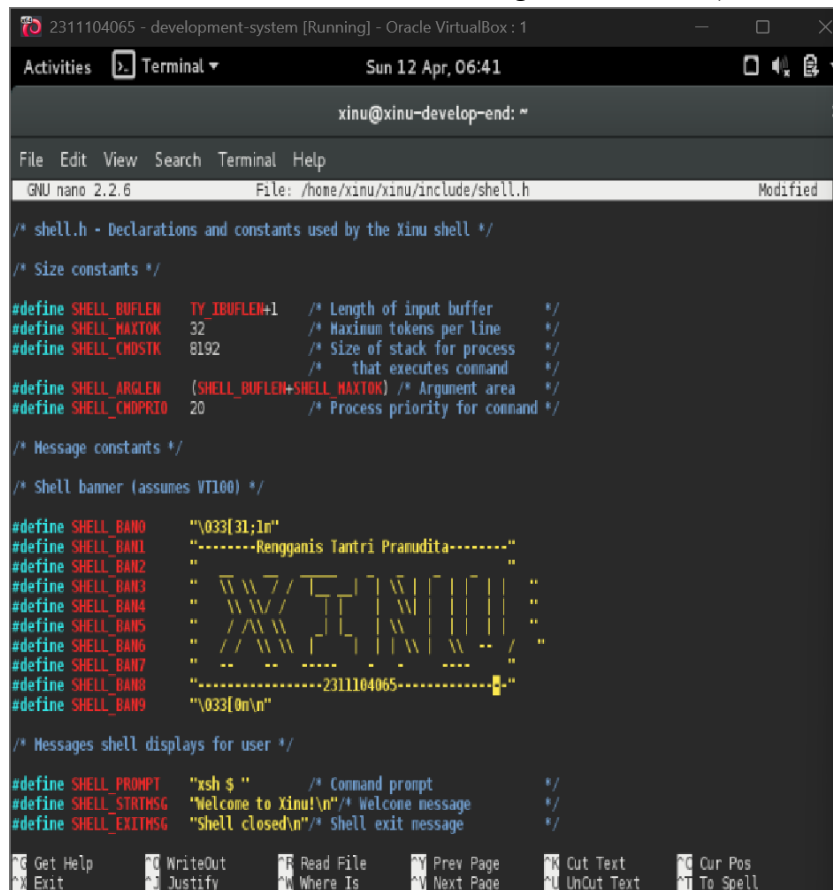
```
xinu@xinu-develop-end: ~  
cat: /home/xinu/xinu/include/home/xinu/xinu/include/shell.h: No such file or directory  
xinu@xinu-develop-end:~$ cat /home/xinu/xinu/include/shell.h  
/* shell.h - Declarations and constants used by the Xinu shell */  
  
/* Size constants */  
#define SHELL_BUFLen TY_IBUFLen+1 /* Length of input buffer */  
#define SHELL_MAXTok 32 /* Maximum tokens per line */  
#define SHELL_OndStk 8192 /* Size of stack for process */  
#define SHELL_OndStk /* that executes command */  
#define SHELL_ArgLen (SHELL_BUFLen+SHELL_MAXTok) /* Argument area */  
#define SHELL_OndPriO 20 /* Process priority for command */  
  
/* Message constants */  
  
/* Shell banner (assumes VT100) */  
#define SHELL_BANd "\033[31;1m"  
#define SHELL_BANd1 "-----"  
#define SHELL_BANd2 "  
#define SHELL_BANd3 "XINU"  
#define SHELL_BANd4 "-----"  
#define SHELL_BANd5 "  
#define SHELL_BANd6 "  
#define SHELL_BANd7 "  
#define SHELL_BANd8 "\033[0m\n"  
  
/* Messages shell displays for user */  
  
#define SHELL_Prompt "xsh $ " /* Command prompt */  
#define SHELL_STrtMSG "Welcome to Xinu!\n" /* Welcome message */  
#define SHELL_ExItMSG "Shell closed\n" /* Shell exit message */  
#define SHELL_SyNerrMSG "Syntax error\n" /* Syntax error message */  
#define SHELL_CrEerrMSG "Cannot create process\n" /* command error */  
#define SHELL_InErMSG "Cannot open file %s for input\n" /* Input err */
```

- Carilah file yang menampilkan banner Xinu! Hint: file berekstensi pada direktori xinu/shell



```
2311104065 - development-system [Running] - Oracle VirtualBox : 1  
xinu@xinu-develop-end: ~  
xinu@xinu-develop-end:~$ clear  
[[3;3]  
xinu@xinu-develop-end:~$ grep -rl "banner\\|Welcome\\|XINU" /home/xinu/xinu/shell/ --include="*.c"  
/home/xinu/xinu/shell/shell.c  
xinu@xinu-develop-end:~$ cat /home/xinu/xinu/shell/shell.c  
/* shell.c - shell */  
  
#include <xinu.h>  
#include <stdio.h>  
#include "shprototypes.h"  
  
/* Table of Xinu shell commands and the function associated with each */  
const struct cmdent cmdtab[] = {  
    {"argcho", xsh_argcho},  
    {"arp", xsh_arp},  
    {"cat", xsh_cat},  
    {"clear", xsh_clear},  
    {"date", xsh_date},  
    {"devdump", xsh_devdump},  
    {"echo", xsh_echo},  
    {"help", xsh_help},  
    {"ls", xsh_ls},  
    {"kill", xsh_kill},  
    {"memdump", xsh_memdump},  
    {"memstat", xsh_memstat},  
    {"ns", xsh_ns},  
    {"netinfo", xsh_netinfo},  
    {"ping", xsh_ping},  
    {"ps", xsh_ps},  
    {"sleep", xsh_sleep},  
    {"tee", xsh_tee},  
    {"udp", xsh_udpdump},  
    {"udpecho", xsh_udpecho},  
    {"udpeserver", xsh_udpeserver},
```


- c. Ubahlah welcome banner Xinu (Tambahkan NAMA dan NIM masing-masing, kemudian ubah banner Xinu sesuai dengan selera Anda)



```
2311104065 - development-system [Running] - Oracle VirtualBox : 1
Sun 12 Apr, 06:41
xinu@xinu-develop-end: ~
File Edit View Search Terminal Help
GNU nano 2.2.6 File: /home/xinu/xinu/include/shell.h Modified

/* shell.h - Declarations and constants used by the Xinu shell */

/* Size constants */

#define SHELL_BUFLN TY_IBUFLN+1 /* Length of input buffer */
#define SHELL_MAXTOK 32 /* Maximum tokens per line */
#define SHELL_CNDSTK 8192 /* Size of stack for process */
/* that executes command */
#define SHELL_ARGLEN (SHELL_BUFLN+SHELL_MAXTOK) /* Argument area */
#define SHELL_CNDPRIO 20 /* Process priority for command */

/* Message constants */

/* Shell banner (assumes VT100) */

#define SHELL_BAN0 "\\033[31;1m"
#define SHELL_BAN1 "-----Rengganis Tantri Pramudita-----"
#define SHELL_BAN2 " "
#define SHELL_BAN3 "  W W /  T  T  W W  |  |  |  |  |  "
#define SHELL_BAN4 "  W W /  T  T  W W  |  |  |  |  |  "
#define SHELL_BAN5 "  / W W  T  T  W W  |  |  |  |  |  "
#define SHELL_BAN6 "  / W W  T  T  W W  |  |  |  |  |  "
#define SHELL_BAN7 "  ----- 2311104065 -----  "
#define SHELL_BAN8 " "
#define SHELL_BAN9 "\\033[0m\\n"

/* Messages shell displays for user */

#define SHELL_PROMPT "xsh $ " /* Command prompt */
#define SHELL_STRMSG "Welcome to Xinu!\\n" /* Welcome message */
#define SHELL_EXITMSG "Shell closed\\n" /* Shell exit message */

Get Help WriteOut Read File Prev Page Cut Text Cur Pos
Exit Justify Where Is Next Page UnCut Text To Spell
```

Ubah baggian SHELL_BAN0 menjadi nama yaitu Rengganis Tantri Pramudita dan baggian SHELL_BAN8 menjadi NIM yaitu 2311104065

- d. Setelah melakukan perubahan lakukan perintah berikut ini pada terminal Linux
- \$ cd xinu/compile
 - \$ make clean
 - \$ make
 - \$ sudo minicom
 - Jalankan vm backend

```

xinu@xinu-develop-end: ~/xinu/compile
File Edit View Search Terminal Help
4447448 bytes of free memory. Free list:
    [0x0015E320 to 0x0009FFFF]
    [0x00100000 to 0x005FBFFF]
116361 bytes of XINU code.
    [0x00100000 to 0x001C0000]
148392 bytes of data.
    [0x001FF20 to 0x001442C7]

Obtained IP address 10.0.3.15   (0xa0a0030f)

-----Rengganis Tantri Pramudita-----

      \X//  _[]_  |[]|  |||| 
       \X//  |[]|  ||||  |||| 
        \X//  |_|_|  ||||  |||| 
         \X//  |_|_|  ||||  |||| 
          \X//  |_|_|  ||||  |||| 
           \X//  |_|_|  ||||  |||| 
            \X//  |_|_|  ||||  |||| 
             \X//  |_|_|  ||||  |||| 
              \X//  |_|_|  ||||  |||| 
               \X//  |_|_|  ||||  |||| 

-----2311104065-----

Welcome to Xinu!

xsh $ S
```